

IndicConformer Marathi Evaluation Summary Report

RESPIN Dialect Corpus Baseline Results

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Evaluation Overview

- **Evaluated Model:** ai4bharat/indicconformer_stt_mr_hybrid_ctc_rnnt_large
- **Dataset:** RESPIN Marathi Sub-Corpus (4 Dialect Groups, 2 Domains)
- **Total Utterances Evaluated:** 2,170 utterances
- **Total Audio Duration:** 3.042 hours

Overall Benchmark Comparison: CTC vs. RNNT

Decoder Mode	Raw WER	Norm WER	Raw CER	Norm CER	Norm Exact Match
CTC	28.46%	23.94%	7.26%	4.95%	30.00%
RNNT	28.54%	23.71%	7.39%	5.06%	30.78%

Dialect-Wise Breakdown (Normalized Metrics)

Code	Dialect Region / Variety	Utts	CTC Norm WER	RNNT Norm WER	RNNT Norm CER
D1	South Konkan (Sindhudurg / Ratnagiri)	559	34.59%	34.37%	7.45%
D2	North Konkan (Palghar / Thane)	540	24.64%	24.14%	5.08%
D3	Standard Pune (Benchmark Target)	555	11.57%	10.87%	2.34%
D4	Varhadi (Vidarbha / Amravati)	516	24.29%	24.79%	5.14%
Total	All Dialects Combined	2,170	23.94%	23.71%	5.06%

Domain-Wise Breakdown (Normalized Metrics)

Domain	Utterances	CTC Norm WER (%)	RNNT Norm WER (%)	RNNT Norm CER (%)
Agriculture	1,060	23.95%	23.85%	5.36%
Banking	1,110	23.94%	23.59%	4.77%

Key Empirical Insights

1. **Prestige Target Baseline (D3 Standard Pune):** The model achieves its best performance on Standard Pune speech (~10.87% WER / 2.34% CER under RNNT), confirming that the ASR backbone is well-calibrated for standard Marathi.
2. **South Konkan (D1) Dialect Gap:** South Konkan exhibits the highest error rate across all dialects (**34.37% WER**), generating a +23.50% **absolute WER gap** compared to Standard Pune. This underscores the severe phonological and acoustic mismatch between coastal Konkani/South Konkan speech and standard ASR acoustic models.

3. **Varhadi (D4) & North Konkan (D2) Discrepancy:** Both Varhadi and North Konkan exhibit $\sim 24.1\%$ – 24.8% WER, demonstrating a consistent $\sim 13.5\%$ absolute WER degradation over standard speech.
4. **CTC vs. RNNT Decoding Parity:** CTC and RNNT decoders perform very similarly overall (23.94% vs 23.71% Normalized WER), though RNNT provides a slight edge on Standard Pune (10.87% vs 11.57%) and Exact Match Accuracy (30.78% vs 30.00%).